Build 02: Acoustic Cymbal Hoverboard Bench Rig

Purpose

This is the larger student build. It should be run as a bench force platform, with no rider and no free-flight claim.

The rig uses cymbals, pot lids, stainless bowls, or similar thin metal dishes as resonant acoustic radiators. Piezo or Langevin transducers drive the dishes. The students map conventional near-field acoustic force first, then use the same apparatus as a larger coherent-body test article.

Primary goal:

Measure force versus frequency, phase, drive amplitude, and standoff distance, then test whether a sign-reversible coherent state produces any residual after conventional acoustic effects are subtracted.

Build Stages

Stage B0: Single-Dish Calibration Station

Build this before the four-dish platform.

Item	First build target
Dish	200 mm aluminum pot lid, stainless bowl, or small cymbal
Transducers	3 to 12 piezo discs or bolt-on transducers
Layout	Ring at about 60 percent radius
Drive	Phase-locked channels from one MCU
Measurement	Suspended dish above separate ground plate on load cell
Standoff	Adjustable 0.5 to 30 mm
Output	Force curves and resonance maps

Correct measurement geometry:

- The driven dish is supported by an overhead force gauge or a frame independent of the ground plate.
- The reflecting ground plate sits on its own load cell or scale.
- Do not put both dish and ground plate on the same scale. Internal action/reaction forces will cancel in the reading.

Stage B1: Four-Dish Bench Platform

Build this after B0 is stable.

Item	First build target
Dishes	4 x 200 to 300 mm stainless bowls or pot lids
Optional larger dishes	4 x 400 to 510 mm cymbals for later work
Transducers	3 per dish, 12 total
Placement	120 degrees apart, about 60 percent radius
Orientation	Bell or convex focus downward, aimed at ground plate
Frame	Square frame, 600 to 800 mm side length for large dishes, smaller for bowls
Tilt	0 degrees for pure vertical force maps. Optional 10 to 15 degrees outward for steering studies
Drive	12 phase-locked channels
Measurement	Overhead load cells, corner load cells, or separate ground-plate load cells
Test surface	Smooth glass, polished aluminum, or polished concrete

Bill Of Materials

Single-dish calibration:

- 1 aluminum pot lid, stainless mixing bowl, or cymbal.
- 3 to 12 piezo or Langevin transducers. Start with lower-power parts.
- M3 or M4 bolts, washers, and rubber gaskets if using bolt-on transducers.
- RP2040/RP2350/Teensy/ESP32-class controller.
- 3 to 12 class-D amplifier channels or MOSFET drive channels.
- Current-limited DC supply for bench calibration.
- Pickup piezo or accelerometer on the dish.
- Load cell plus HX711 or better ADC, or a lab scale with serial logging.
- Adjustable standoff fixture with micrometer, screw jack, or spacer set.
- Acrylic shield or other fragment guard.
- Remote kill switch.
- Hearing protection.

Four-dish platform:

- 4 matching dishes.
- 12 transducers.
- Lightweight nonmagnetic frame.
- 12 drive channels, current-limited.
- MCU with phase-locked waveform generation.
- Per-dish pickup sensor or accelerometer.
- Battery pack for later self-contained runs.
- Temperature sensors on transducers and dishes.
- Load cells at support points or a separate ground-plate reaction measurement.
- Shielding, remote kill switch, and acoustic enclosure if available.

Mechanical Build

1. Mark each dish at 60 percent radius.
2. Place three transducers 120 degrees apart on each dish.
3. Avoid the rim and exact center. The rim is fragile and the center is a poor first location for many modes.
4. If drilling thin metal, use a center punch and step drill. Clamp the dish gently to avoid deformation.
5. Use rubber gaskets for bolt holes where acoustic sealing matters.
6. Route wires along the dish and frame. Strain-relieve every cable.
7. Add a pickup piezo or accelerometer to each dish.
8. Put a temperature sensor near at least one transducer per dish.
9. Build the standoff fixture so the gap to the ground plate can be set repeatably.
10. Add a transparent shield before high-amplitude sweeps.

Drive Architecture

Minimum:

- one MCU generates phase-locked PWM or DAC waveforms,
- one amplifier channel per transducer,
- one pickup sensor per dish,
- frequency sweep and resonance-lock mode,
- fixed state schedules written to the log.

Drive states:

State	Behavior
`OFF`	No drive. Sensors and logger only.
`SHAM`	Same average power, incoherent or phase-scrambled drive.
`ACTIVE_PLUS`	Coherent multichord drive with intended sign.
`ACTIVE_MINUS`	Same power, reversed phase gradient or dish zoning.

State	Behavior
`DUMMY`	Matched non-coherent dish or mechanically damped dish.

Start with one frequency at a time. Add multichord drive only after the single-frequency force curves are understood.

Measurement Plan

Conventional Acoustic Force Map

For each dish:

1. Set standoff distance: 0.5, 1, 2, 5, 10, 20, and 30 mm.
2. Sweep frequency around measured resonances.
3. Sweep drive amplitude at fixed frequency.
4. Record dish acceleration, transducer temperature, load-cell force, supply voltage, current, and acoustic level if available.
5. Repeat on different surfaces: glass, aluminum, polished concrete, and a lossy surface such as foam or carpet.

Expected conventional signatures:

- force changes sharply with standoff,
- force peaks at resonances,
- force grows with drive amplitude,
- force weakens on lossy surfaces,
- force follows thermal and acoustic leakage when artifacts dominate.

Four-Dish Platform Map

After each dish has a force curve:

1. Drive all four dishes with the same amplitude and phase pattern.
2. Measure total vertical reaction.
3. Drive one dish at a time to calibrate corner contributions.
4. Test phase patterns that should cancel lateral components.
5. Test differential drive only as a steering-force measurement, not as a rider control.
6. Compare `ACTIVE_PLUS`, `ACTIVE_MINUS`, and `SHAM` at matched average power.

Claim Boundary

This rig can demonstrate conventional near-field acoustic force. That measurement is a valid student result.

A human-carrying hoverboard is outside scope. The source material itself contains a correction: human-scale free-air acoustic hover is blocked by acoustic intensity, wavelength, and safety limits, and the realistic cymbal/pot-lid first-light result is small bench-scale deflection.

Treat any residual force claim as provisional until:

- the conventional force map is complete,
- the residual survives standoff changes,
- the residual reverses with `ACTIVE_MINUS`,
- the residual reverses under physical flip or rotation,
- the matched dummy rejects it,
- thermal, electrostatic, magnetic, acoustic, and vibration artifacts are rejected.

Safety

- No rider, no standing on the platform, no hands between dish and ground plate.
- Start at low power and short duty cycles.
- Keep a hard kill switch within reach.
- Use hearing protection. Harmonics can be loud even when the drive frequency seems harmless.
- Shield the rig. Cymbals, bowls, and piezos can crack under overdrive.

- Stop if any transducer reaches 60 C.
- Use current-limited supplies.
- Do not run high-amplitude tests near loose objects, dust, paper, or uncovered sensors.
- Log every crack, deformation, desoldered joint, or temperature excursion.

Student Deliverables

- Single-dish force curves.
- Resonance map for each dish.
- Standoff-force map.
- Surface-dependence map.
- Four-dish total force curve.
- Raw logs and analysis scripts.
- Safety log.
- Residual or upper-bound report.